

KEYNOTES

PALEOCLIMATES & PALEOENVIRONMENTS



UTILIZING A NETWORK OF ARCTICA ISLANDICA RECORDS AND MODEL SIMULATIONS TO INVESTIGATE PAST REGIONAL OCEAN DYNAMICS IN THE RAPIDLY WARMING WESTERN NORTH ATLANTIC

The complex changes occurring in modern day oceans require an understanding of long-term regional processes to contextualize current observations and guide future projections. As sclerochronology continues to grow in the field of paleoclimatology, the ability to build networks of shell-based, hydrographic reconstructions has become critical to understanding complex oceanic regions in the past. This comprehension is particularly critical in areas like the rapidly warming western North Atlantic, where the confluence of multiple major ocean currents and water masses create a complex hydrodynamic system. This presentation will discuss efforts to build a network of six shell-based hydrographic reconstructions on the continental shelf off the eastern coast of the United States, ranging from the southern Mid-Atlantic Bight to the northern Gulf of Maine. Reconstructions utilize annually- to decadal-resolved oxygen isotope records from *Arctica islandica* shells and vary in length from ~60 years to over 300 years. At some sites, radiocarbon and nitrogen isotope records have also been developed. Reconstructions reveal both coherent warming within the last ~50 years as well as deviations in variability between sites farther back in time, likely indicating multiple oceanographic processes affecting these different regions. Such variability was further investigated using a hierarchy of numerical model simulations, including a fully coupled, global climate model (Community Earth System Model – Last Millennium Ensemble) and a high-resolution ocean model (VIKING20X). Combining sclerochronological records with model simulations enables a more thorough investigation and interpretation of past regional oceanic changes in order to contextualize the rapid changes observed today.

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